23~24 微甲Ⅱ小测2：

1. 设 $f(x,y)$ 在 R 上具有连续偏导，$f(1,1)=1$，$f_x'(1,1)=1$，$f_y'(1,1)=3$. 记 $\varphi(x)=f(x, f(2x-1,x^3))$，则 $\varphi'(1)=$ 34 (87.1%)

解：$\left.\frac{d\varphi(x)}{dx}\right|_{x=1} = f_x'(x, f(2x-1,x^3)) + f_y'(x, f(2x-1,x^3))$

$\left[2f_x'(2x-1,x^3) + 3x^2 f_y'(2x-1,x^3)\right]\Big|_{x=1}$

$= f_x'(1,1) + f_y'(1,1)\left[2f_x'(1,1) + 3f_y'(1,1)\right]$

$= 1 + 3\times(2+3\times 3) = 34.$

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2. 曲线 $C: \begin{cases} x^2+y^2+2z^2=10 \\ z=2xy \end{cases}$ 在点 $(1,1,2)$ 处的切线方程为 (AC).

A. $\frac{x-1}{1} = \frac{y-1}{-1} = \frac{z-2}{0}$.

B. $\frac{x-1}{1} = \frac{y-1}{-1} = \frac{z-2}{1}$.

C. $\begin{cases} x+y+4z=10 \\ x+y+2z-6=0 \end{cases}$

D. $\begin{cases} x+2y+z=6 \\ 2x-2y-z+2=0 \end{cases}$

E. 若其他没法选，则选此项.

解：$\vec{v} = \vec{n_1}\times\vec{n_2}$

$= (F_x', F_y', F_z')\times(G_x', G_y', G_z')$

$= (2x, 2y, 4z)\times(-2y, -2x, 1)\Big|_{(1,1,2)}$

$= \begin{vmatrix} i & j & k \\ 2 & 2 & 8 \\ -1 & -1 & 1 \end{vmatrix} = 10i - 10j$

$\Rightarrow (1,-1,0)$ 为 A.

验证C与D：C符合；D不行.

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3. 设 $z=f(x,y)$ 在 $(1,2)$ 处有一阶连续偏导，且沿 $\vec{u}=\{3,4\}$，$\vec{v}=\{4,-3\}$ 的方向导为 $\frac{\partial f}{\partial \vec{u}}=18$，$\frac{\partial f}{\partial \vec{v}}=-1$，则 $z=f(x,y)$ 在 $(1,2)$ 处的全微分为：$dz = 10dx + 15dy$. (76.5%)

解：$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$

$\Rightarrow \begin{cases} \frac{1}{5}\left(3\frac{\partial f}{\partial x} + 4\frac{\partial f}{\partial y}\right) = 18 \\ \frac{1}{5}\left(4\frac{\partial f}{\partial x} - 3\frac{\partial f}{\partial y}\right) = -1 \end{cases}$ (归一化) $\Rightarrow \begin{cases} \frac{\partial f}{\partial x} = 10 \\ \frac{\partial f}{\partial y} = 15 \end{cases}$ $(1,2)$

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4. $u=\sqrt{x^2+2y^2+3z^2}$ 在 $M(-2,1,-1)$ 处最大方向导数为：$\frac{1}{3}\sqrt{17}$ (82.6%)

解：$\frac{\partial u}{\partial \vec{v}} = \frac{\partial u}{\partial x}\cos\alpha + \frac{\partial u}{\partial y}\cos\beta + \frac{\partial u}{\partial z}\cos\gamma \overset{(-2,1,-1)}{=\!=\!=} -\frac{2}{3}\cos\alpha + \frac{2}{3}\cos\beta - \cos\gamma.$

设为 a,b,c，则 $L = (-2a+2b-3c)\frac{1}{3} + \lambda(a^2+b^2+c^2-1)$

4. 解：$\begin{cases} \frac{\partial L}{\partial a} = -\frac{2}{3} + 2a\lambda = 0 \\ \frac{\partial L}{\partial b} = \frac{2}{3} + 2b\lambda = 0 \\ \frac{\partial L}{\partial c} = -1 + 2c\lambda = 0 \\ \frac{\partial L}{\partial \lambda} = a^2+b^2+c^2-1 = 0 \end{cases} \Rightarrow \begin{cases} a = \frac{1}{3\lambda} \\ b = -\frac{1}{3\lambda} \\ c = \frac{1}{2\lambda} \end{cases} \Rightarrow \lambda = \pm\frac{\sqrt{17}}{6}$.

∴ 最大为 $\frac{1}{3}(-2a+2b-3c) = \frac{1}{3}\sqrt{17}$.

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5. 设 $f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, (x,y) \neq (0,0) \\ 0, (x,y) = (0,0) \end{cases}$，则下列陈述正确的是：(BD).

A. $f(x,y)$ 沿任何方向的方向导为零.

B. $f(x,y)$ 在 $(0,0)$ 处连续.

C. $f'_x(0,0) = f'_y(0,0) = 0$，且 $f(x,y)$ 在 $(0,0)$ 处可微.

D. $f'_x(0,0) = f'_y(0,0) = 0$，但 $f(x,y)$ 在 $(0,0)$ 处不可微.

E. 若其他选项都没法选，则选此项.

解：B：① $\lim\limits_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = f(0,0)$.

$\because xy \le \frac{x^2+y^2}{2}$ $\therefore 0 \le \left|\frac{xy}{\sqrt{x^2+y^2}}\right| \le \frac{1}{2}\sqrt{x^2+y^2} = \frac{\rho}{2}$

$\rho \to 0$，夹逼，则 $f(x,y) \to 0$ $((x,y)\to(0,0))$. ✓.

② $f'_x(0,0) = \lim\limits_{x\to 0} \frac{f(x,0)-f(0,0)}{x} = \lim\limits_{x\to 0} 0 = 0 = f'_y(0,0)$.

③ $\lim\limits_{\rho\to 0} \frac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\rho} = \lim\limits_{\rho\to 0} \frac{xy}{\rho^2} = 0$ 可微.

∴ C：✓，D：✗.

④ 方向导：$\frac{\partial f}{\partial \vec{v}} = \lim\limits_{\rho\to 0} \frac{f(\rho\cos\theta, \rho\sin\theta)-f(0,0)}{\rho} = \lim\limits_{\rho\to 0} \frac{\frac{\rho^2}{2}\sin 2\theta}{\rho^2} = \frac{1}{2}\sin 2\theta$

不恒为 0. A ✗.

6、若 $f(x,y)$ 在点 $P(1,-2)$ 处连续，且满足：

$$f(x,y)=5+x-y+x^2+4(x-1)(y+2)+(y+2)^2+o((x-1)^2+(y+2)^2)\quad(x\to1,y\to-2).$$

则：下列结论中正确的是：（BD）. 70.78%.

A. $f_x'(1,-2)=1$.　　B. $df|_{(1,-2)}=3dx-dy$.

C. $f(x,y)$ 在 $(1,-2)$ 处有极值.

D. 曲面 $z=f(x,y)$ 在 $(1,-2,f(1,-2))$ 处切面为 $z=3x-y+4$.

E. 若其他选项都没法选.

解：A. $f_x'(1,-2)=\lim\limits_{x\to0}\dfrac{f(1+x,-2)-f(1,-2)}{x}=\lim\limits_{x\to0}\dfrac{x^2+3x+o(x^2)}{x}=3$. ×

B. $f_y'(1,-2)=\lim\limits_{y\to0}\dfrac{f(1,-2+y)-f(1,-2)}{x}=\lim\limits_{y\to0}\dfrac{y^2-y+o(y^2)}{y}=-1$. √

C. 一阶导不为0，故无极值. D. 对. E. ×

7、设 $u(x,y)$ 在有界闭区域 D 上具有二阶连续偏导，且满足：

$\dfrac{\partial^2u}{\partial x\partial y}\neq0$，$\dfrac{\partial^2u}{\partial x^2}+\dfrac{\partial^2u}{\partial y^2}=0$，则下列正确的是（B）. (87.74%)

A. u 的最大值点与最小值点必在 D 内.

B. u 的最大值点与最小值点必在 D 边界上.

C. u 的最大值点在 D 内部，最小值点在 D 的边界上.

D. u 的最小值点在 D 的内部，最大值点在 D 边界上.

E. 若其他选项都没法选，则选此项.

8、已知 $D=\{(x,y)\mid x^2+y^2\le2\}$，则 $\iint\limits_D(2x^2-3xy+4y^2)\,dxdy=$ 6π.

(73.38%)

解：原式 $=\int_0^{\sqrt2}dr\int_0^{2\pi}(2r^2\cos^2\theta-3r^2\sin\theta\cos\theta+4r^2\sin^2\theta)\,r\,d\theta$.

$=\int_0^{\sqrt2}r^3dr\int_0^{2\pi}(3-\cos2\theta-\frac{3}{2}\sin2\theta)\,d\theta$

$=6\pi$.

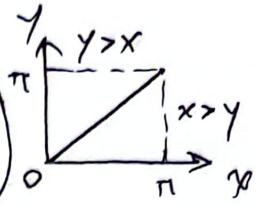
9. 设 $D=\{(x,y)\mid 0\le x\le\pi,\ 0\le y\le\pi\}$，则 $\iint_D \sin(\max\{x,y\})\,dxdy$

$=$ ____2π____. (86.27%)

解：原式 $=\int_0^\pi dx\int_0^x \sin x\,dy+\int_0^\pi dx\int_x^\pi \sin y\,dy$.

$=\int_0^\pi(-\cos x+1)\,dx+\int_0^\pi(\cos x+1)\,dx$

$=\pi+\pi=2\pi$.



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10. 设平面区域 D 是由曲线 $y=\sin\frac{\pi}{2}x$ 与直线 $y=1$，$x=-1$ 围成，则二重积分 $\iint_D(12x^2\sqrt{1-y^2}+x^3\sin y)\,d\sigma=$ ____2π____. (56.86%).

解：原式 $=\int_{-1}^1 dx\int_{\sin\frac{\pi}{2}x}^1(12x^2\sqrt{1-y^2}+x^3\sin y)\,dy$.

$=\int_{-1}^1 dy\int_{-1}^{\frac{2}{\pi}\arcsin y}(12x^2\sqrt{1-y^2}+x^3\sin y)\,dx$.

$\because\int\sqrt{1-y^2}\,dy=\frac{1}{2}(\arcsin y+y\sqrt{1-y^2})+C$.

$\therefore$ 原式 $=\int_{-1}^1 dx\left[6x^2(\arcsin y+y\sqrt{1-y^2})\right]\Big|_{\sin\frac{\pi}{2}x}^{1}$

$+\int_{-1}^1 dy\left[\frac{1}{4}x^4\sin y\right]\Big|_{-1}^{\frac{2}{\pi}\arcsin y}$

（对称、偶、奇）

$=\int_{-1}^1 6x^2\left(\arcsin 1+0-\frac{\pi}{2}x-\sin\frac{\pi}{2}x\cos\frac{\pi}{2}x\right)dx$

$=\int_{-1}^1 6x^2\left[\frac{\pi}{2}-\frac{\pi}{2}x-\frac{1}{2}\sin\pi x\right]dx$

$=\left[\pi x^3-\frac{3}{4}\pi x^4\right]\Big|_{-1}^{1}-\int_{-1}^1 3x^2\sin\pi x\,dx$

（对称、偶、奇）

$=2\pi$.